Case Study: The Future of Work

Forecasting Global Unemployment Disparities (1991–2025)

1. Case Study Overview

Industry: Public Policy / Economic Research
Business Context: Global economic stability is closely linked to employment rates. International organizations like the International Labour Organization (ILO) require predictive insights to identify regions or demographic groups that need targeted financial aid or labor market interventions.

2. The Business Problem

Despite global economic growth, unemployment rates remain volatile due to events like financial crises, pandemics, and automation. The primary challenge is to predict future unemployment rates (specifically for 2024 and 2025) based on historical trends (1991–2023) across 183 different countries, accounting for gender and age disparities.

3. Dataset Description

The dataset (disoccupazione.csv) contains 57,519 records across the following dimensions:

- Country / ISO Code: 183 unique nations.
- Year: Historical data from 1991 to projections for 2025.
- Sex: Segments for Total, Male, and Female.
- Age Groups: 15+, 15-24 (Youth), and 25+.
- Observed Value (obs_value): The unemployment rate (%) recorded or projected.

Data Visualizations

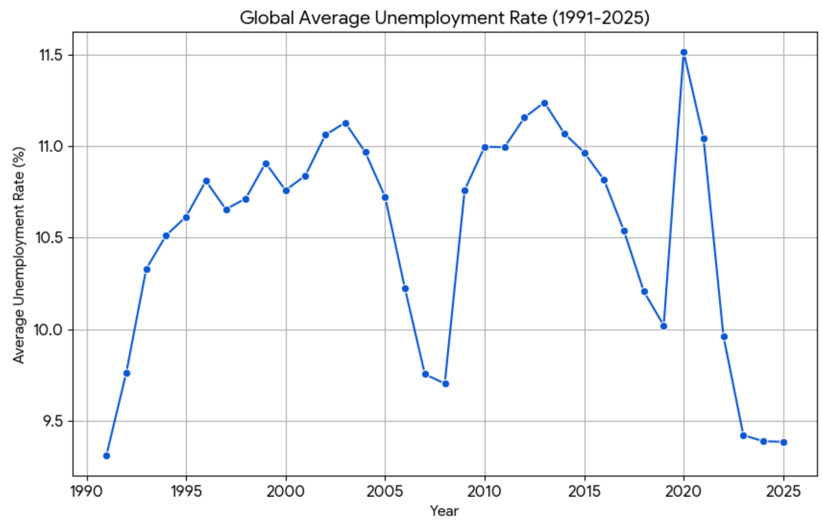

Figure 1: Global Average Unemployment Trend (1991-2025)

Figure 2: Global Youth Unemployment Trend by Gender

4. Project Objectives

- Regression/Forecasting: Predict the obs_value for the year 2025 for specific country-demographic pairs.
- Risk Classification: Identify "High-Risk Countries" where unemployment is predicted to exceed 20%.
- Trend Analysis: Quantify the "Gender Gap" and predict its trajectory.

5. Technical Roadmap

Phase 1: Exploratory Data Analysis (EDA)

Analyze global trends, identify regional hotspots (e.g., Southern Africa), and measure demographic disparities.

Phase 2: Feature Engineering

Create lag features (previous year's rate), calculate rolling averages, and encode categorical variables like country and age group.

Phase 3: Modeling & Evaluation

Utilize models such as XGBoost or Random Forest. Use a Time-Series Split for validation (Train: 1991-2022, Test: 2023-2025).

6. Potential Business Impact

- Early Warning Systems: Proactive job training for youth in high-risk zones.
- Resource Allocation: Prioritizing aid for countries with rising unemployment projections.
- Gender Policy: Data-driven justification for female empowerment initiatives.